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PROJECT OF HOME AUTOMATION IOT BASED

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ABSTRACT

This project focuses on developing an IoT-based Home Automation System that utilizes internet connectivity to automate and control common household appliances. Using a Wi-Fi-enabled ESP32 microcontroller, the system allows remote access through a mobile application. Key functionalities include automatic water tank management, light control based on sunlight, remote fan operation, automated plant irrigation, and complete home power control — all enabled and monitored over the internet.



INTRODUCTION

Home automation is the integration of technology into everyday life to simplify and automate household tasks. The system we developed uses Internet of Things (IoT) technology to allow users to control and monitor their home appliances remotely via the internet.

The core of the system is the ESP32 microcontroller, which connects to Wi-Fi and enables communication between the physical appliances and the Blynk mobile app. With this, users can turn appliances on or off, check sensor readings, and manage their home even while away.

This project demonstrates practical implementations such as automatic water filling, light control based on sunlight, mobile-controlled fans, and automatic plant irrigation. These operations are not only automated but also internet-enabled, giving users full control anytime, anywhere.

PROJECT OVERVIEW

This smart home system is built around Internet of Things (IoT) technology. All modules are linked through the internet to provide real-time automation:

Water Tank System: Automatically operates based on sensor data and can be monitored remotely.

Light Control: Works automatically and can also be overridden via mobile using the internet.

Fan Control: Can be turned ON/OFF using the Blynk app via Wi-Fi.

Irrigation System: Waters plants automatically and notifies the user over the internet.

Main Power Cut-off: Internet-based remote switching of the entire home supply adds safety during electrical issues.

HARDWARE COMPONENTS

1) ESP 32 MICRO CONTROLLERS
2) ULTRASONIC SENSOR
3) IRRIGATION SENSOR
4) LDR SENSOR
5) RELAY MODULE
6) LED
7) WATER PUMP AND MINI WATER PUMP
8) JUMPER WIRES
9) BREADBOARD

SOFTWARE COMPONENTS

1) ARDUINO IDE
2) BLYNK MOBILE APPLICATION

TECHNOLOGIES

Internet of Things (IoT)

Main Foundation of the project.

IoT allows devices (sensors, controllers, and actuators) to communicate with each other over the internet.

ESP32 connects to Wi-Fi and shares data with the mobile app (Blynk cloud).

Enables remote monitoring and control.

Embedded Systems

The system is built using microcontrollers (ESP32) that are programmed to control hardware based on sensor input.

Embedded C/C++ is used to write the logic inside the controller.

Wireless Communication (Wi-Fi)

ESP32 has built-in Wi-Fi module, which connects the entire system to the internet.

All remote operations (turning fan ON/OFF, checking water level, etc.) happen through this wireless link.

Sensor Technologies

Ultrasonic Sensor: For water level detection.

LDR Sensor: For automatic light sensing.

Soil Moisture Sensor: For checking soil condition in plant pots.

These sensors collect environmental data and send it to the ESP32.

Relay Switching

Relays act as electrically operated switches.

ESP32 sends signals to the relays to turn appliances ON/OFF (like fans, lights, water pump).

Ensures isolation and safety between microcontroller and high voltage appliances.

Cloud Computing (Blynk Cloud Platform)

Blynk provides the cloud infrastructure that connects the ESP32 and mobile application.

It stores sensor data temporarily and allows command transmission over the internet.

Provides real-time dashboards and control panels on mobile.

Mobile Application (Blynk App)

User interface to control devices and monitor sensors.

Interacts with Blynk cloud over the internet.

Works on both Android and iOS.

Provides buttons, sliders, and data displays.

Arduino IDE (Software Platform)

Used to write and upload code to ESP32.

Open-source development environment.

Supports embedded C/C++ programming with ESP32 libraries.

MODULE

1) Automatic Water Tank Control Module
2) Automatic Lighting Control Module
3) Fan Control Module
4) Automatic Irrigation System for Indoor Plants
5) Main Power Supply Control Module
6) Central Controller – ESP32 Module

MODULE DESCRIPTION

This IoT-based Home Automation Project is composed of the following key modules, each playing a critical role in automating different household functions. All modules are integrated with the ESP32 microcontroller, which connects them to the internet via Wi-Fi, enabling remote monitoring and control through a mobile application (Blynk).

Automatic Water Tank Control Module

Objective: To automatically manage water filling in the overhead tank and prevent overflow or dry run of the motor.

Sensor Used: Ultrasonic Sensor

How It Works:

The ultrasonic sensor is placed on top of the tank to measure the water level.

When water level drops below 25%, the ESP32 triggers the water pump ON.

When the tank is full, the sensor detects it and automatically turns the motor OFF.

Role of Internet:

Real-time tank level status is sent to the user via the Blynk app.

The motor can also be controlled remotely through the internet in case of manual override.

Automatic Lighting Control Module

Objective: To turn home lights ON/OFF automatically based on ambient light conditions.

Sensor Used: LDR (Light Dependent Resistor)

How It Works:

The LDR sensor continuously measures light intensity.

During daytime, if light is sufficient, lights remain OFF automatically.

At night or in low light, the ESP32 triggers the lights to turn ON automatically.

Role of Internet:

Light status is visible on the mobile app.

User can remotely control lights via the internet, overriding automatic operation if needed.

Fan Control Module

Objective: To operate home fans from anywhere using a mobile device.

Component Used: Relay module for switching + Blynk App

How It Works:

The fan is connected to a relay module, which is controlled by the ESP32.

Through the Blynk app connected via Wi-Fi, the user can turn the fan ON/OFF remotely.

Role of Internet:

Allows remote operation from any part of the world as long as the ESP32 and mobile are connected to the internet.

Provides feedback of fan state (ON/OFF) in real-time.

Automatic Irrigation System for Indoor Plants

Objective: To keep indoor plants watered automatically even when the user is not at home.

Sensor Used: Soil Moisture Sensor

How It Works:

The sensor checks moisture levels in the soil.

If the soil is dry, the ESP32 activates a small water pump to irrigate the plant.

Once the soil is moist enough, the pump is automatically turned OFF.

Role of Internet:

User receives notifications about soil status via the Blynk app.

Manual control of the irrigation system is also possible over the internet.

Main Power Supply Control Module

Objective: To protect home appliances by remotely cutting off the main electricity supply in case of voltage fluctuations or emergencies.

Component Used: Relay Module

How It Works:

A relay is installed at the main power line controlled by ESP32.

Through the mobile app, users can cut or restore power remotely.

Role of Internet:

This remote kill switch can be activated from anywhere via Wi-Fi.

Prevents electrical damage when users are not physically present at home.

Central Controller – ESP32 Module

Objective: To act as the core of the system, connecting all modules and enabling internet access.

Features:

Built-in Wi-Fi for cloud communication.

Receives data from sensors and controls appliances through relays.

Communicates with the Blynk cloud server, enabling mobile app integration.

Role of Internet:

Without the internet, remote access and live status monitoring would not be possible.

It makes the project a true IoT solution, allowing smart and remote operation.

PROBLEM OF TRADITONAL HOME SYSTEM

Traditional home systems require manual operation, often leading to energy waste and appliance damage. There's no way to control or monitor devices remotely. This project addresses the problem using an internet-based solution, enabling users to automate and monitor their homes even when they are away.

SOLUTION

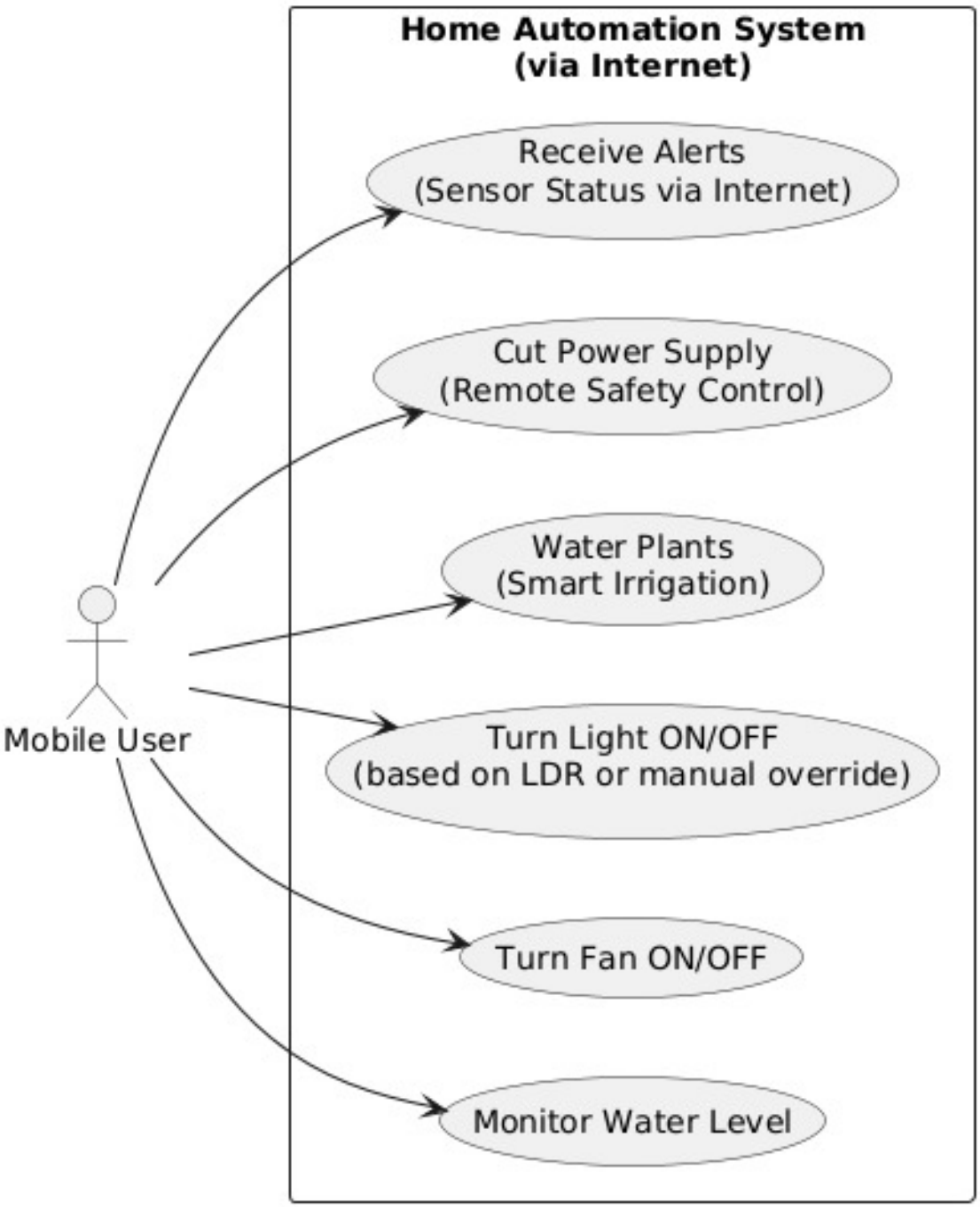
The solution leverages IoT and internet communication to allow remote access and automation. Using the ESP32 and Blynk platform:

The user can monitor sensor readings via the internet.

Devices can be controlled from anywhere using Wi-Fi.

The system ensures automatic responses based on real-time internet-connected sensors.

USE CASE DIAGRAM



CODE

```
#define BLYNK_TEMPLATE_ID "TMPL3vvIjfwTz"

#define BLYNK_TEMPLATE_NAME "HOMEAUTOMATION"

#define BLYNK_AUTH_TOKEN "A76-O4ZGwWD24HooaxQj9atZdHIAYP27"

#define BLYNK_PRINT Serial

#include <WiFi.h>

#include <BlynkSimpleEsp32.h>

char auth[] = BLYNK_AUTH_TOKEN;

char ssid[] = "Abhi";  // type your wifi name

char pass[] = "12341234";  // type your wifi password

#define TRIG_PIN 5

#define ECHO_PIN 18

#define RELAY_PIN1 16 // water tank

#define DO_PIN 13  //LDR sensor

#define RELAY_PIN2 2 // automatice light

#define SOIL_PIN 36 //SP PIN IN ESP32, IRREGATION A0

#define RELAY_PINn1 4 //irregation pump

const int moistureThreshold = 40; // soil moisture threshold for triggering the pump

const int RELAY_PINn2 = 12; // Fan system

const int RELAY_PINn3 = 0;

long duration;

float distance;
```

```
void setup()
 {
  Serial.begin(921600);
  pinMode(DO_PIN, INPUT);
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(RELAY_PIN1, OUTPUT);
  pinMode(RELAY_PIN2, OUTPUT);
  pinMode(RELAY_PINn1, OUTPUT);
  // fan
  Blynk.begin(auth, ssid, pass);
  pinMode(RELAY_PINn2,OUTPUT);
  pinMode(RELAY_PINn3,OUTPUT);
  }

void loop() {
  // Send ultrasonic pulse
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);
```

```
// Read echo time
 duration = pulseIn(ECHO_PIN, HIGH);

 // Calculate distance in cm
 distance = duration * 0.034 / 2;

 Serial.println("Distance: ");
 Serial.print(distance);
 Serial.print(" cm");

 // If water level is low (distance is high), turn on pump
 if (distance < 4)
  {
   digitalWrite(RELAY_PIN1, HIGH);  // Pump ON
   Serial.println("Pump ON");
  }
 else if(distance >7)
  {
   digitalWrite(RELAY_PIN1, LOW);   // Pump OFF
   Serial.println("Pump OFF");
  }
```

```
// automatice light
  int lightState = digitalRead(DO_PIN);


  if (lightState == HIGH){
   digitalWrite(RELAY_PIN2, LOW);
   Serial.println("It is dark");
  }
  else
  {
   digitalWrite(RELAY_PIN2, HIGH);
   Serial.println("It is light");
  }
  //irregation system
  int soilValue = analogRead(SOIL_PIN); // Read soil moisture level (analog value)
  int soilMoisture = map(soilValue, 4095, 0, 0, 100); // Map soil value to
percentage (adjust based on sensor calibration)

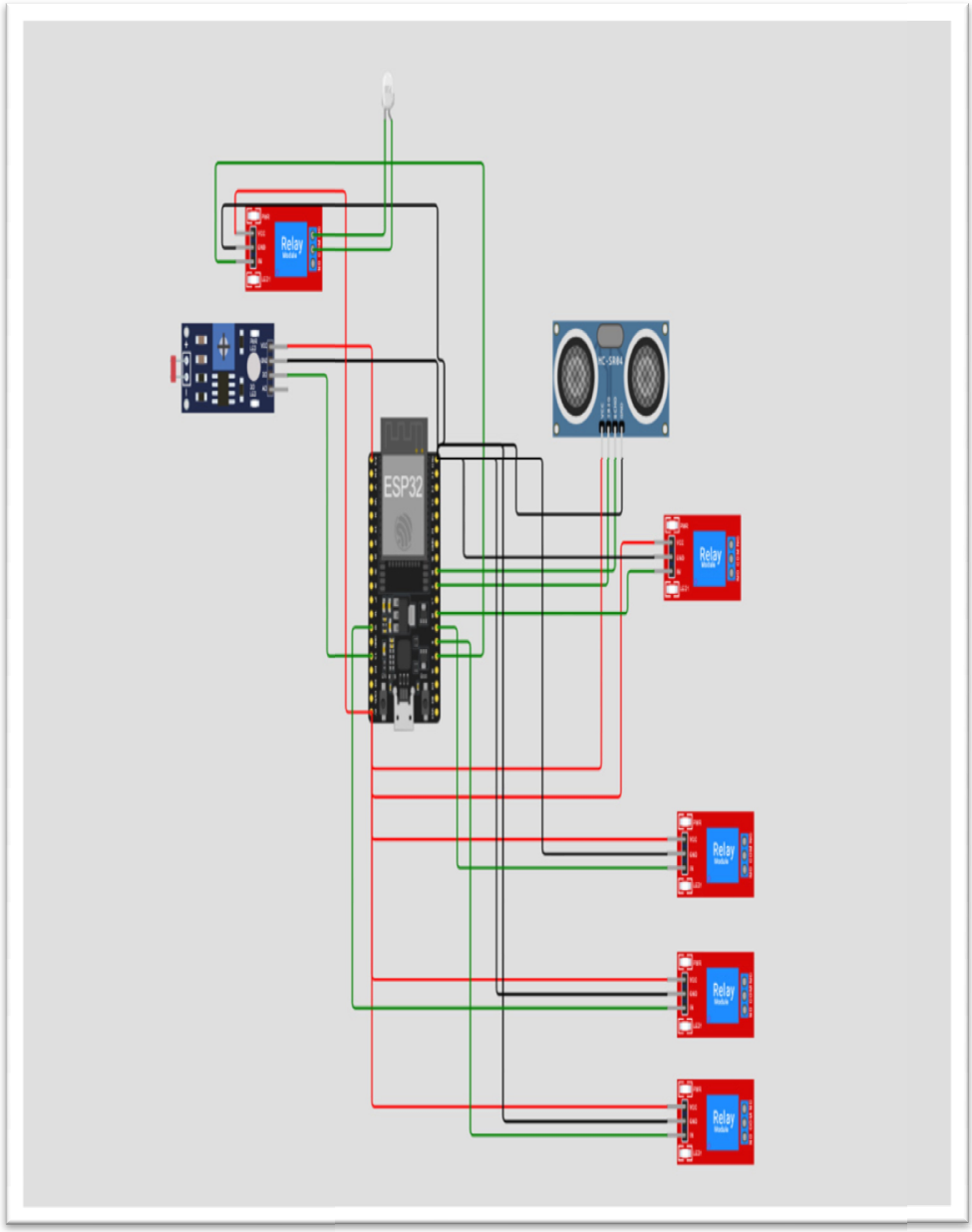

  // Check soil moisture and control pump
  if (soilMoisture < moistureThreshold)
   {
     digitalWrite(RELAY_PINn1, LOW); // Turn on the pump
     Serial.println("Pump is ON - Watering...");
   }
```

```
else
 {
   digitalWrite(RELAY_PINn1, HIGH); // Ensure pump is off
   Serial.println("Pump is OFF - Soil Moisture OK");

 }
  delay(1000);

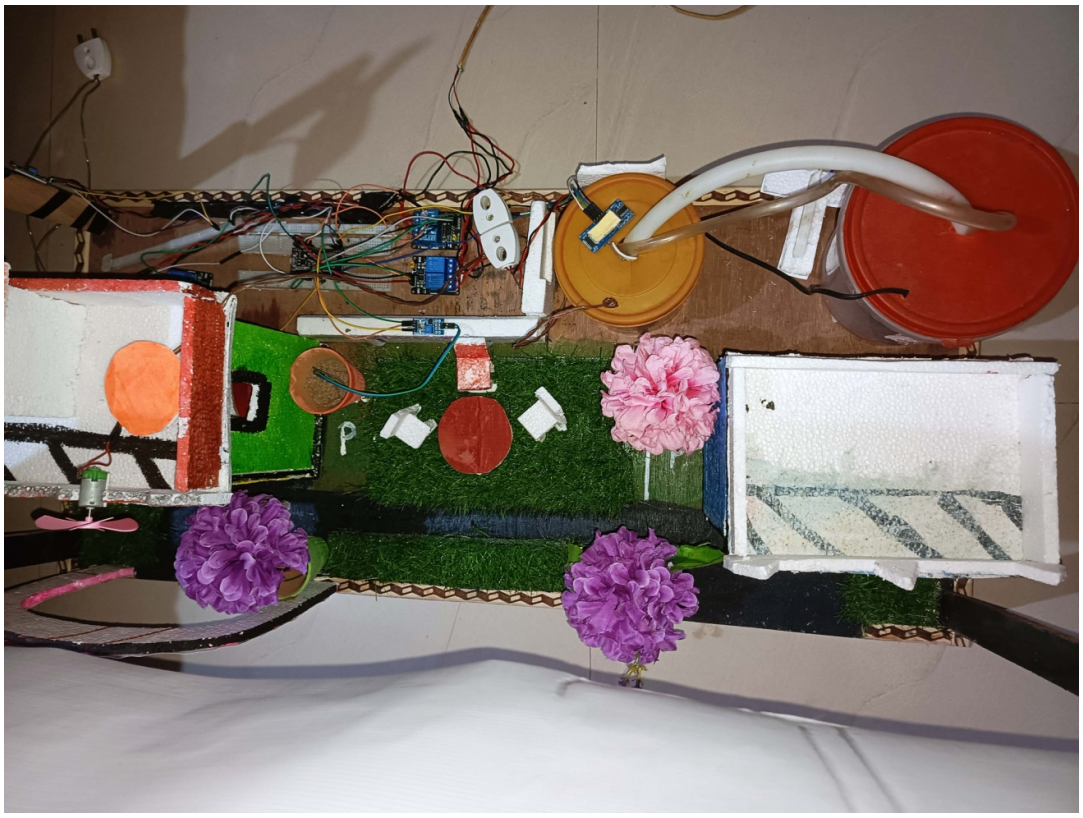
 Blynk.run();
```

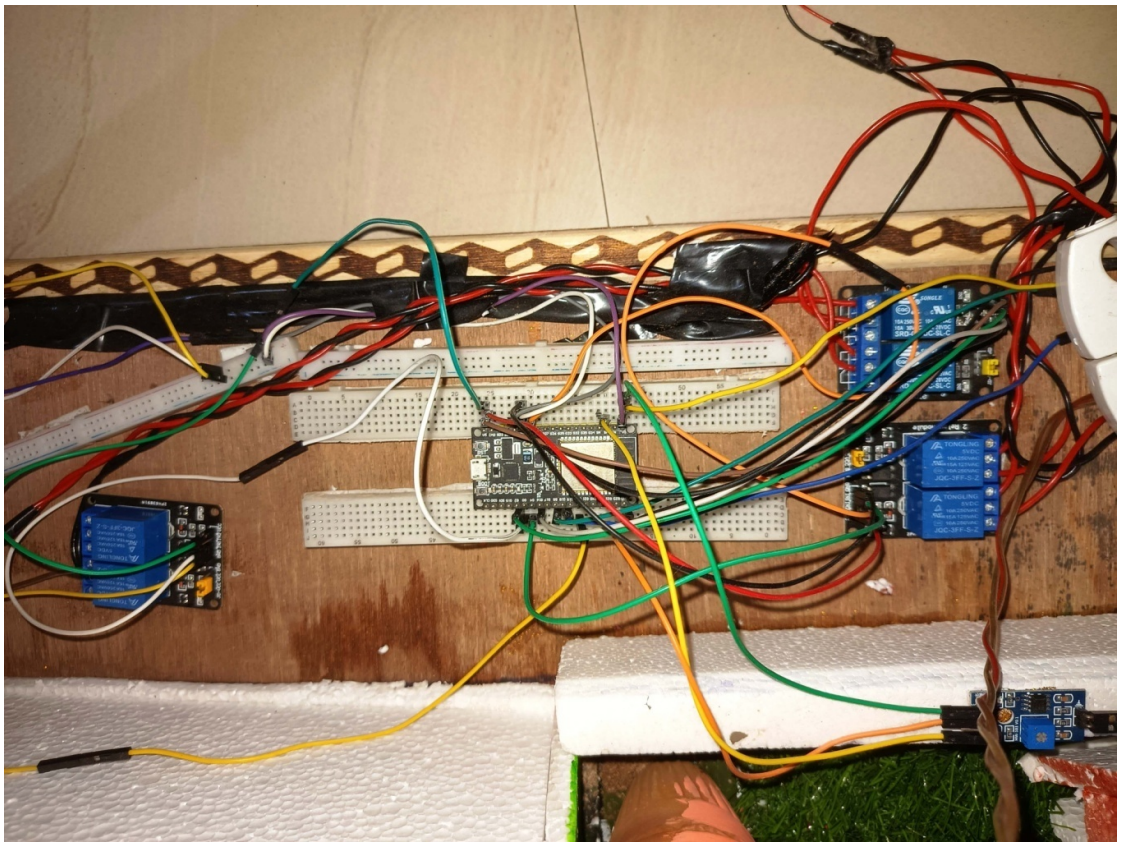
CONNECTIVITY DIAGRAM



SNAPSHOT


GOVT. POLYTECHNIC COLLEGE NOWGONG
PROJECT TITLE - HOME AUTOMATION
Batch-2022-25
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FUTURE SCOPE

Integration with cloud storage for logging data

Control via voice assistants (Google Home, Alexa) over the internet

Enhanced security systems using cameras and live streaming

OTA (Over The Air) updates to devices via Wi-Fi

Expansion to full smart-home ecosystems using cloud APIs and IoT protocols

ESTIMATE OF PROJECT IN HOMEAUTOMATION

S. N.	PRODUCT NAME	PURCHASE PLATFORM	PRICE
1	ESP 32	AMAZON	450
2	ULTRASONIC SENSOR	FLIPKART	179
3	LDR SENSOR	FLIPKART	109
4	IRRIGATION SENSOR (SOIL MOISTURE)	FLIPKART	168
5	JUMBER WIRES	AMAZON	100
6	BREADBOARD	AMAZON	80
7	WATER PUMP OR MINI WATER PUMP	LOCAL MARKET AND FLIPKART	120+60
8	REALY MODULE	LOCAL MARKET AND AMAZON	6*120
9	LED	LOCAL MARKET	40
10	DC MOTOR	LOCAL MARKET	20
11	OTHER PRODUCT	LOCAL MARKET	600
	TOTAL AMOUT		2646

CONCLUSION

This Home Automation System demonstrates how internet-based IoT technology can improve home management. With real-time control, automation, and safety features, the system offers reliability, energy efficiency, and modern convenience. It bridges the gap between physical appliances and digital control, paving the way for smart living.